Module 1:
Introduction, Spatial Neighbors, and Join Test

STAT 679: Statistical Methods for Spatial Lattice Data-Spring 2026
University of Wisconsin-Madison

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March 26, 2026

Outline

1. Lattice Data-From Neighbors to Statistics
2. Introduction to Lattice Data
3. BW Joins/Test I
4. R-Demo
5. Assignment 1

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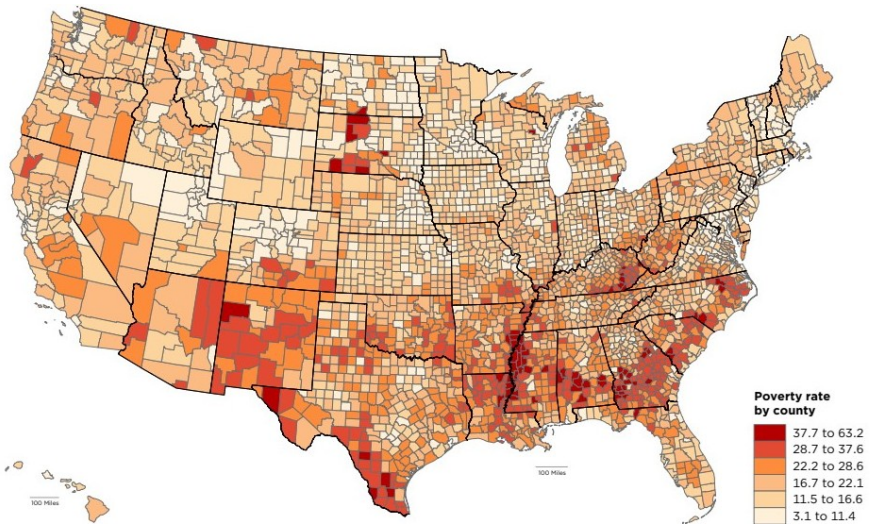
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- Different types of spatial data lead to somewhat different goals and analysis methods.

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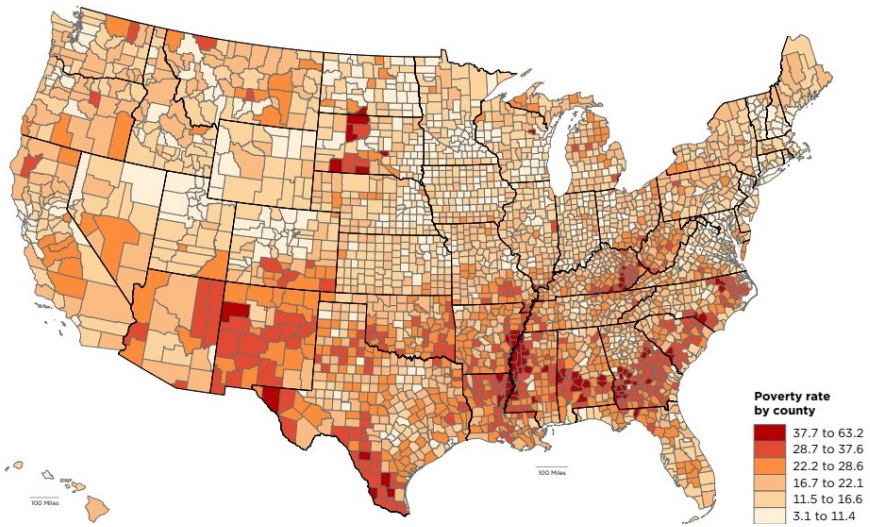
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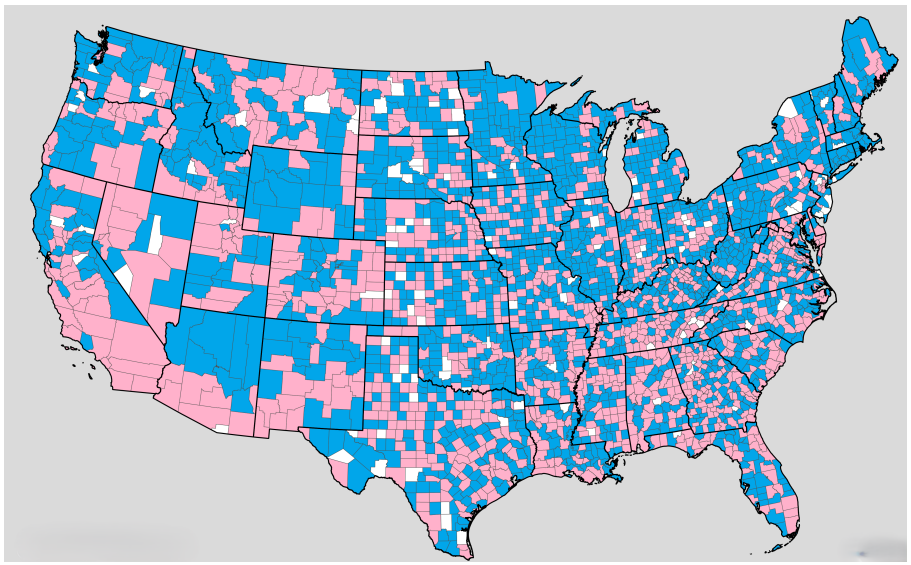


(a) Poverty map by counties

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(a) Poverty map by counties



(b) Male-Female employment by counties

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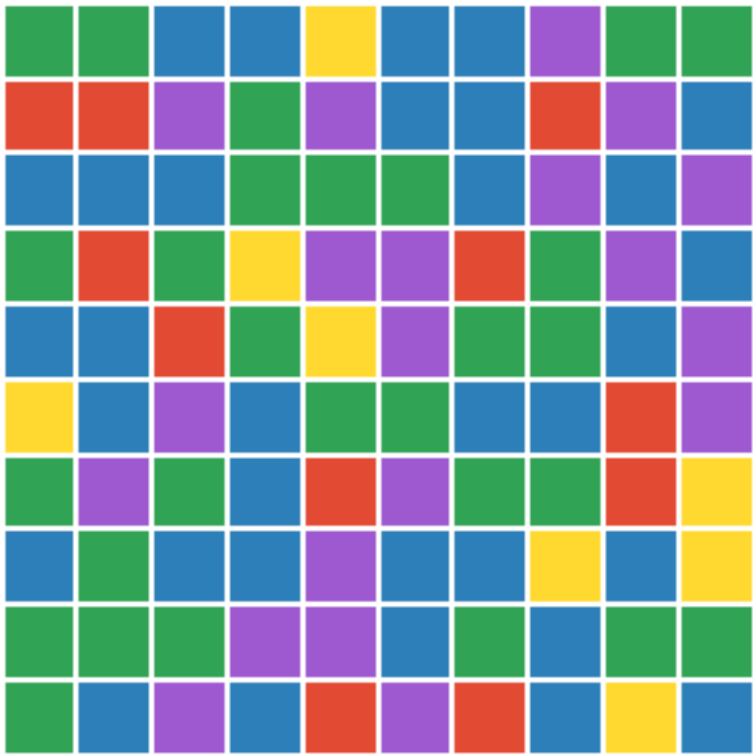
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- Lattice data may be correlated across space.
- Main goals of analysis include:
  - summarizing or quantifying spatial correlation through relationships among adjacent neighbors
  - drawing inference in the presence of spatial correlation

Data Example: Mosaic Pattern

(a) Actual Mosaic



(b) Mosaic Data

G	G	B	B	Y	B	B	P	G	G
R	R	P	G	P	B	B	R	P	B
B	B	B	G	G	G	B	P	B	P
G	R	G	Y	P	P	R	G	P	B
B	B	R	G	Y	P	G	G	B	P
Y	B	P	B	G	G	B	B	R	P
G	P	G	B	R	P	G	G	R	Y
B	G	B	B	P	B	B	Y	B	Y
G	G	G	P	P	B	G	B	G	G
G	B	P	B	R	P	R	B	Y	B

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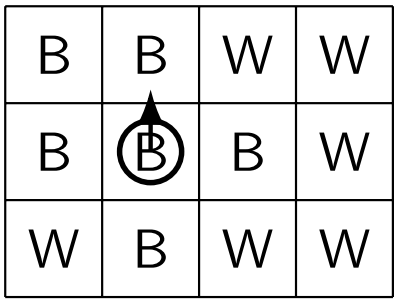
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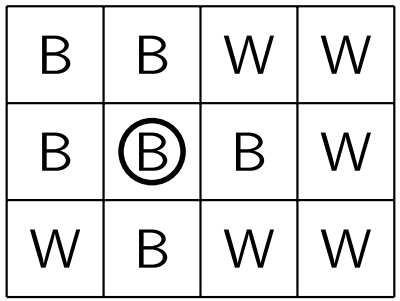
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- Which pattern appears random, regular, or clustered?

Types of Neighbors: Sample Mosaic Example II

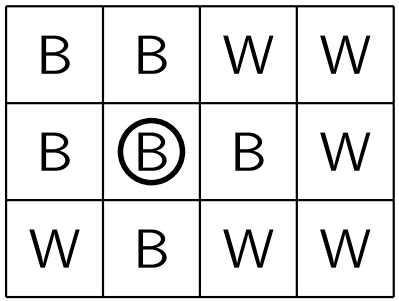
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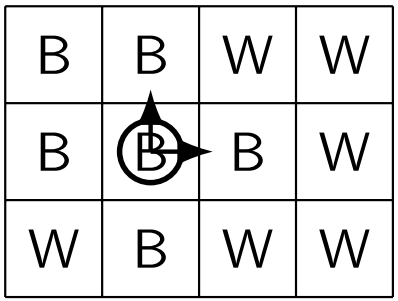
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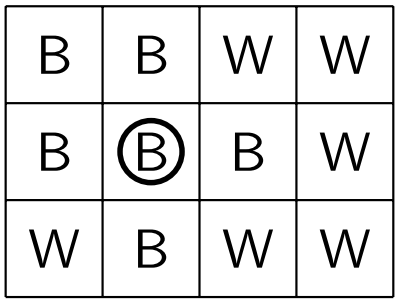
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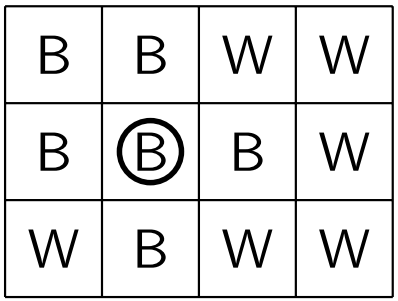
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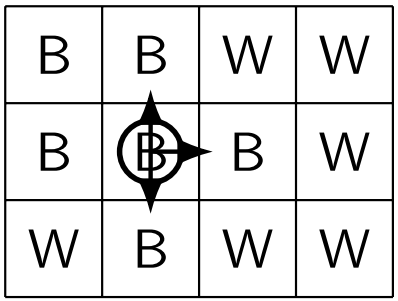
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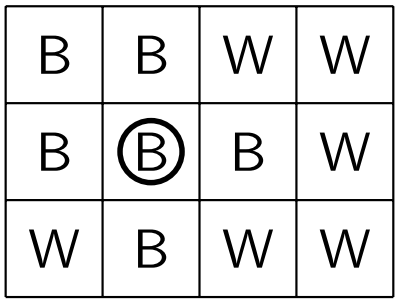
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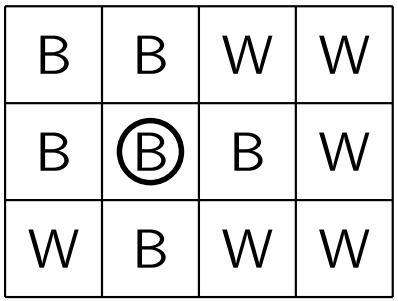
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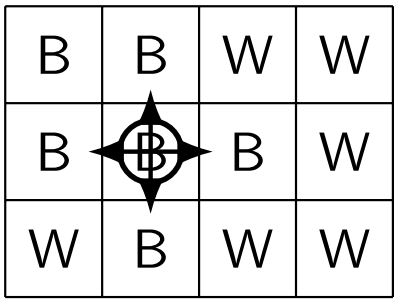
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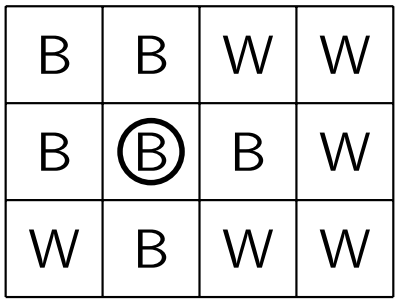
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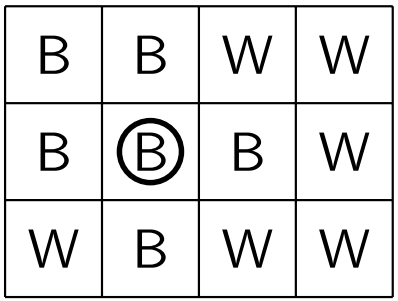
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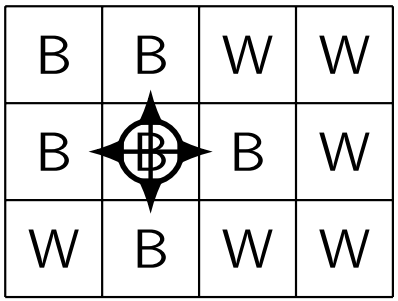
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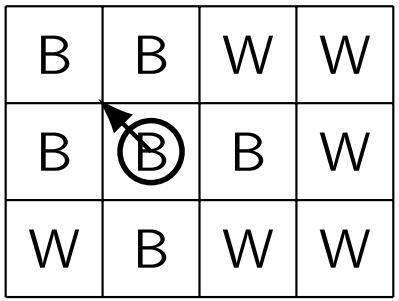
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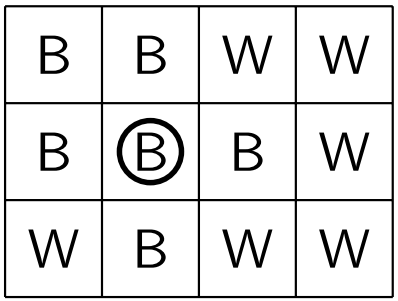
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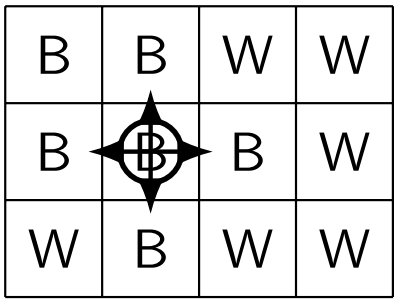
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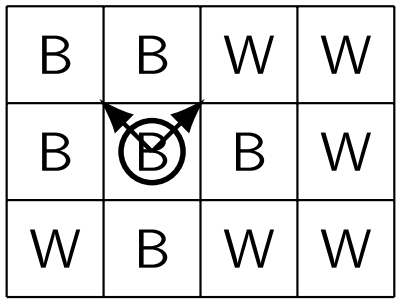
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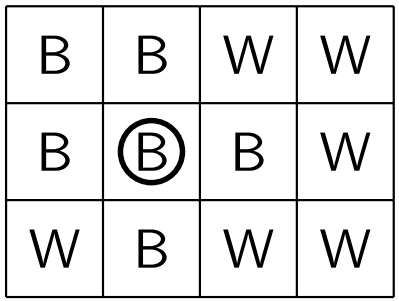
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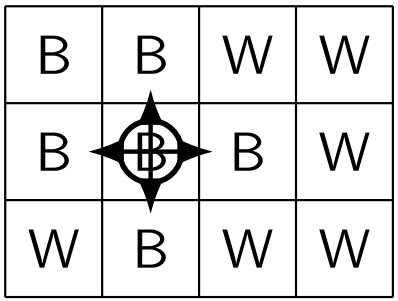
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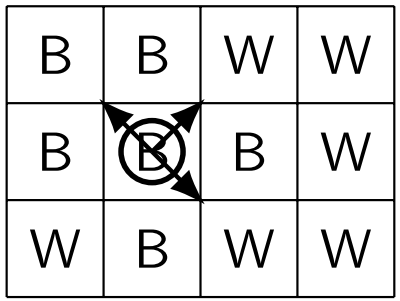
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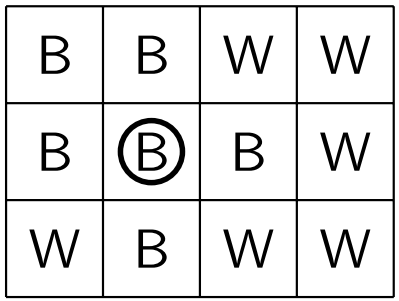
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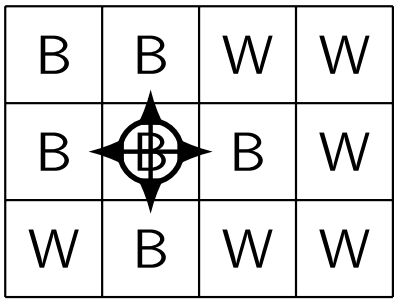
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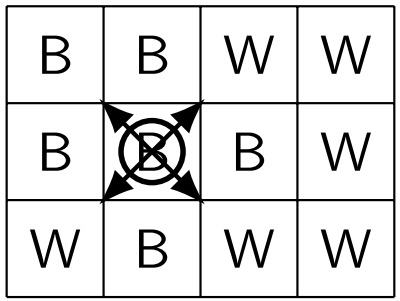
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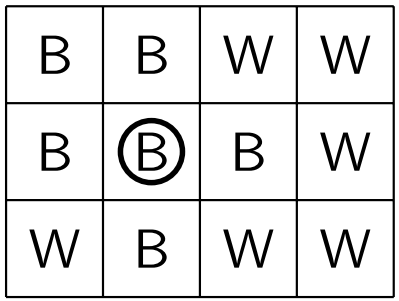
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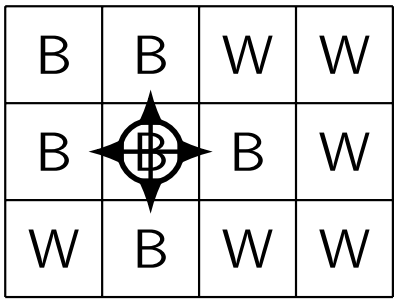
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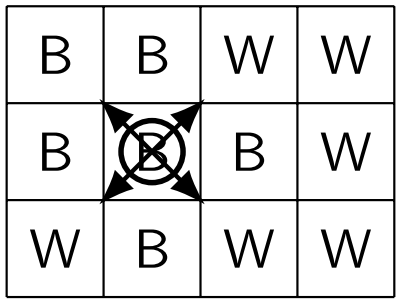
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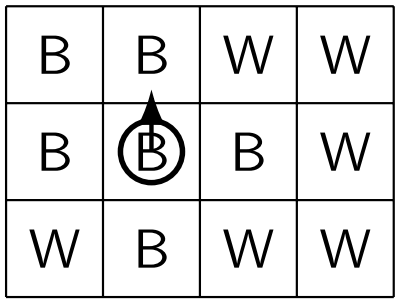
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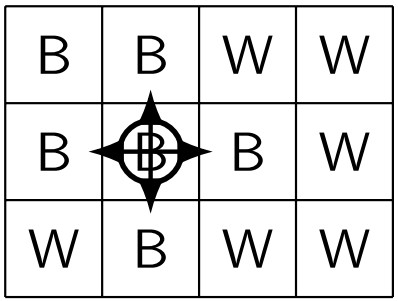
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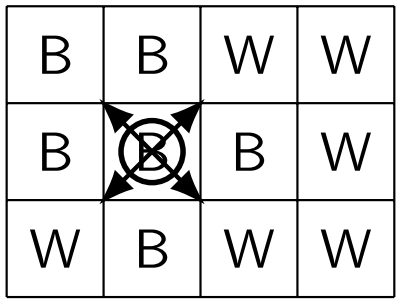
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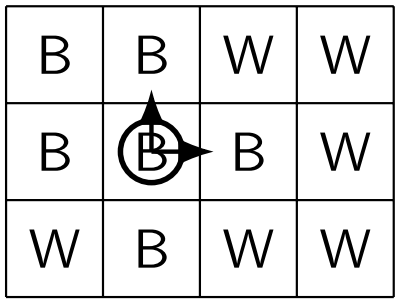
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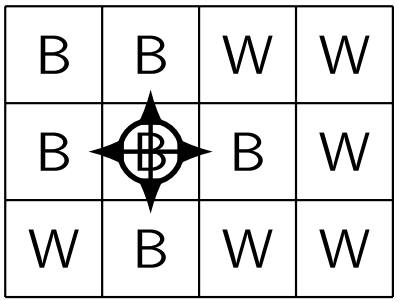
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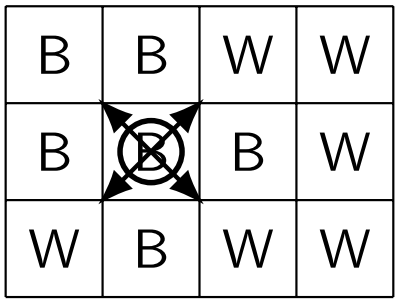
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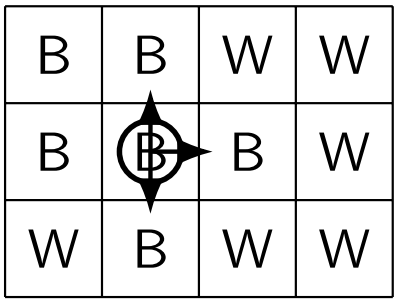
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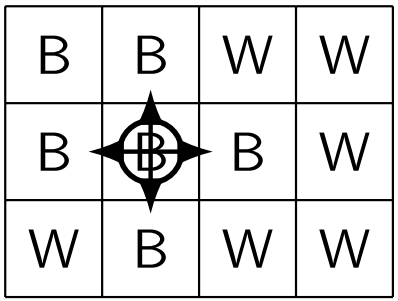
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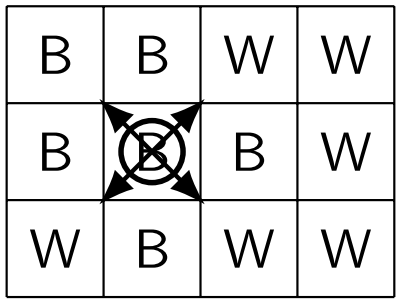
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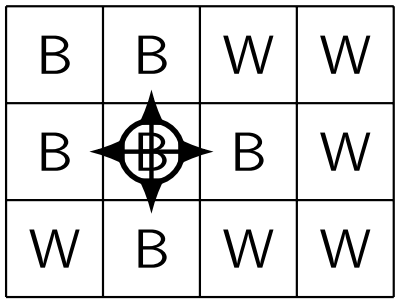
Types of Neighbors: Sample Mosaic Example II



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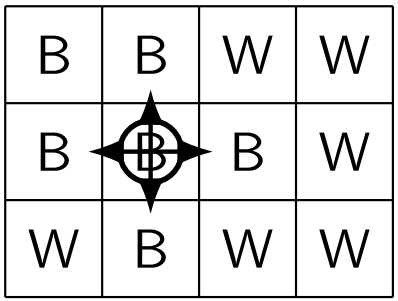
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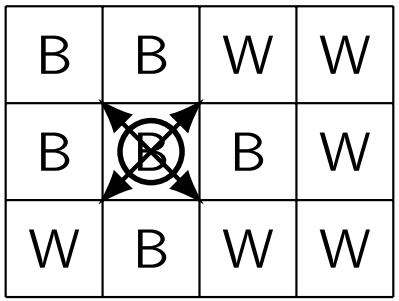
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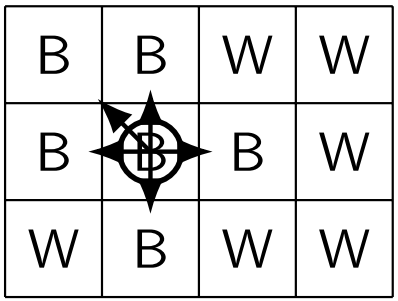
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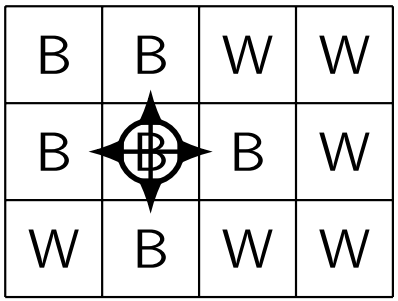
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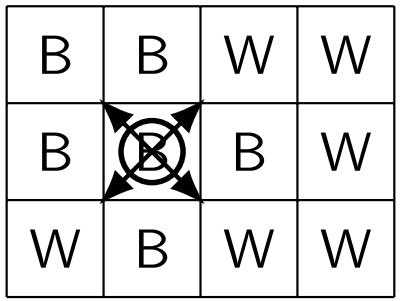
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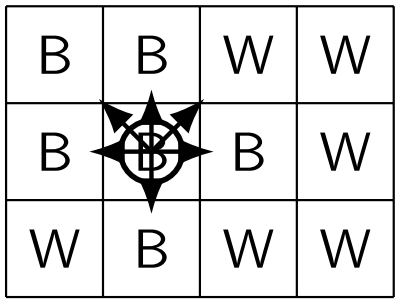
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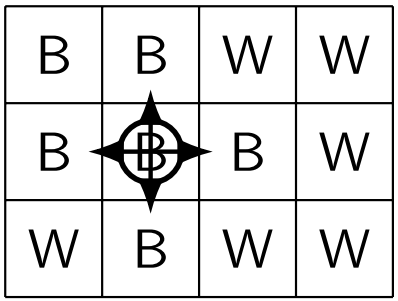
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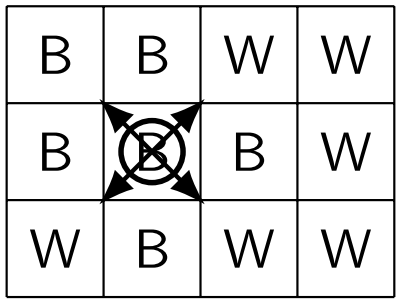
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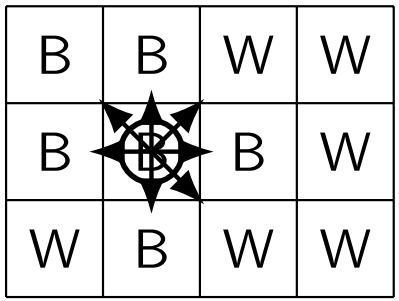
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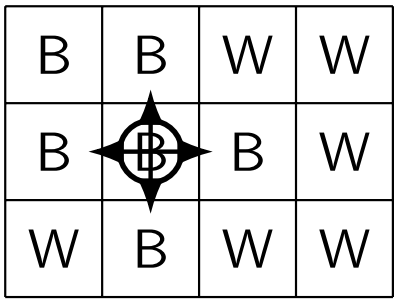
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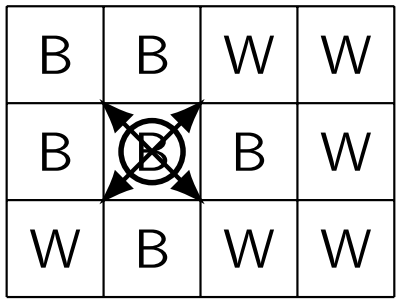
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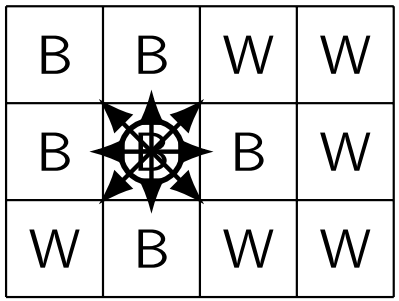
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1. Lattice Data-From Neighbors to Statistics
2. Introduction to Lattice Data
3. BW Joins/Test I
4. R-Demo
5. Assignment 1

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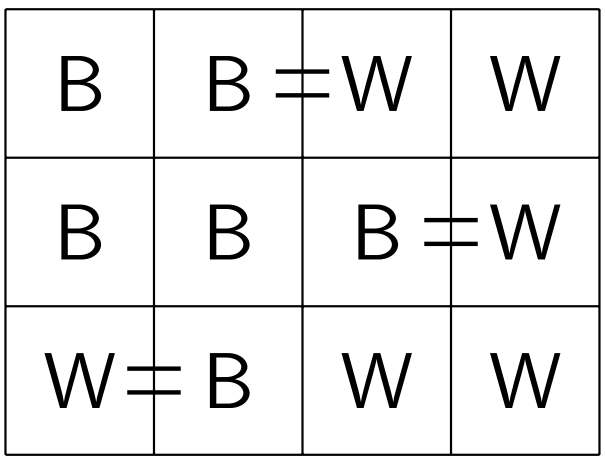
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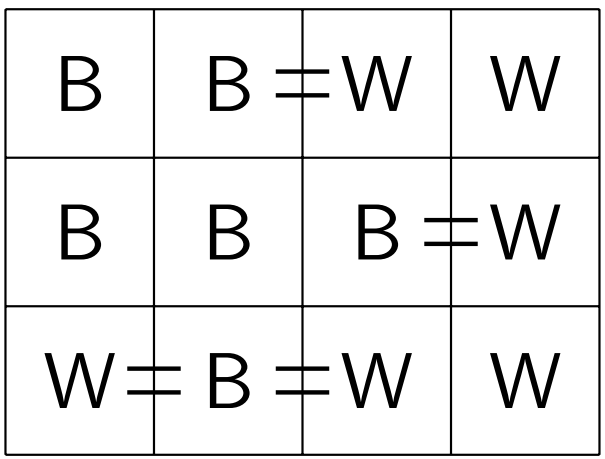
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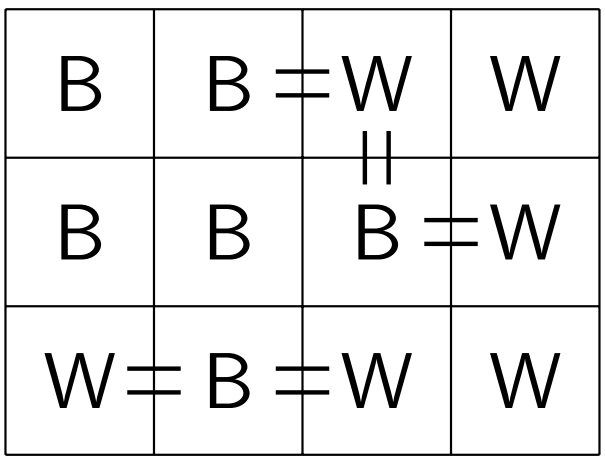
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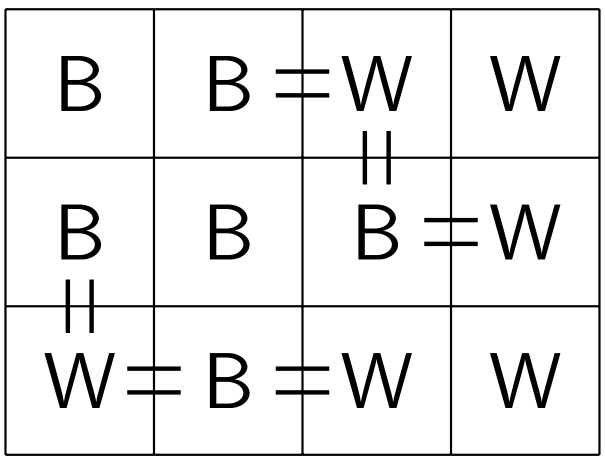
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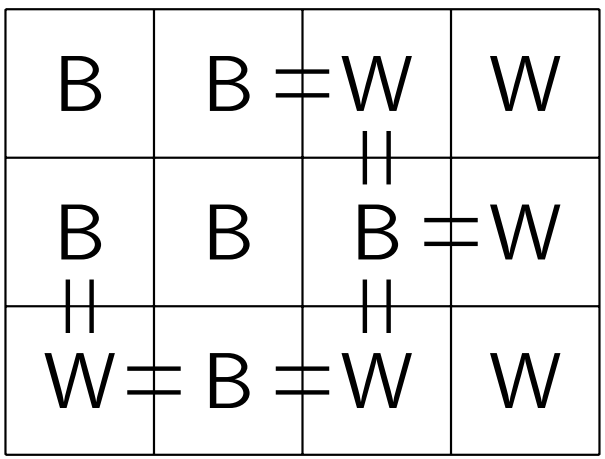
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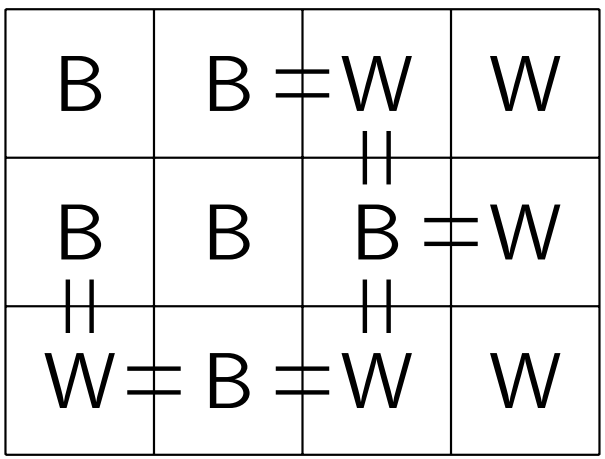
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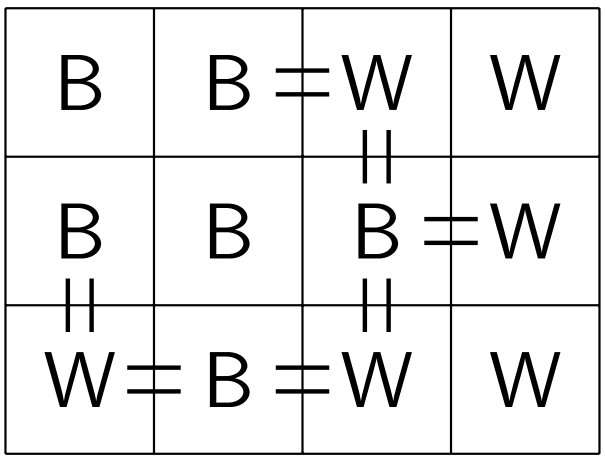
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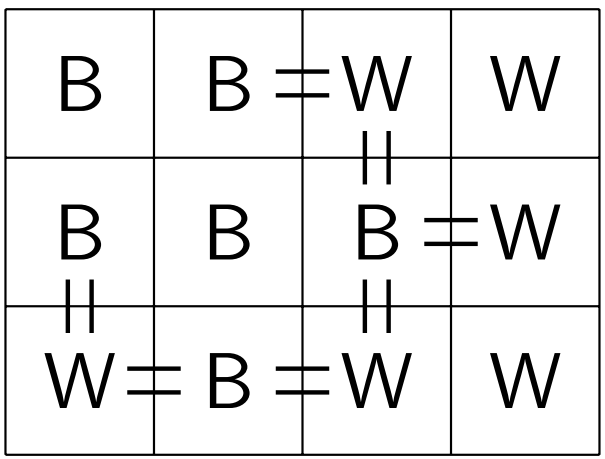
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- Where

$$S_0 = 2(2rc - r - c), \quad S_1 = 2S_0, \quad S_2 = 8(8rc - 7r - 7c + 4),$$

$$r = \#\text{ rows}, \quad c = \#\text{ columns},$$

$$T_0 = 2bw, \quad T_1 = 2T_0, \quad T_2 = 4nbw,$$

$$b = \#\text{ black}, \quad w = \#\text{ white}, \quad n = b + w.$$

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2. Introduction to Lattice Data
3. BW Joins/Test I
4. R-Demo
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R Demo: Mosaic Example

Mosaic Example II

B	B	W	W
B	B	B	W
W	B	W	W

Numeric Coding

1	1	0	0
1	1	1	0
0	1	0	0

Coding scheme: $B = 1,\ W = 0$

R Demo: Load Packages and Enter Data

R code

```
library(spdep)
library(geoR)

# Entering data vector (white = 0, black = 1)
egII <- c(1, 1, 0, 0,
          1, 1, 1, 0,
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Console output

```
     [,1] [,2] [,3] [,4]
[1,]    1    1    0    0
[2,]    1    1    1    0
[3,]    0    1    0    0
```

R Demo: Create Rook Neighborhood Structure

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Console output

```
Neighbour list object:
Number of regions: 12
Number of nonzero links: 34
Percentage nonzero weights: 23.61111
Average number of links: 2.833333
```

R Demo: Cell Numbering and Neighbor Interpretation

R code

```
str(egII.nb)

matrix(seq(1:12), ncol = 4, byrow = TRUE)
```

R Demo: Cell Numbering and Neighbor Interpretation

R code

```
str(egII.nb)

matrix(seq(1:12), ncol = 4, byrow = TRUE)
```

Key console output

```
List of 12
 $ : int [1:2] 2 5
 $ : int [1:3] 1 3 6
 $ : int [1:3] 2 4 7
 ...
 $ : int [1:2] 8 11
```

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```

```
     [,1] [,2] [,3] [,4]
[1,]    1    2    3    4
[2,]    5    6    7    8
[3,]    9   10   11   12
```

R Demo: Summarize the Rook Neighbor List

R code

```
summary(egII.nb)
```

R Demo: Summarize the Rook Neighbor List

R code

```
summary(egII.nb)
```

Console output

```
Neighbour list object:
Number of regions: 12
Number of nonzero links: 34
Percentage nonzero weights: 23.61111
Average number of links: 2.833333

Link number distribution:
2 3 4
4 6 2

4 least connected regions:
1:1 4:1 1:3 4:3 with 2 links

2 most connected regions:
2:2 3:2 with 4 links
```

R Demo: Full Code

```
library(spdep)
library(geoR)
egII = c(1,1,0,0,1,1,1,0,0,1,0,0)
matrix(egII, ncol = 4, byrow = TRUE)

egII.nb = cell2nb(nrow = 3, ncol = 4, type = "rook")
egII.nb

str(egII.nb)
matrix(seq(1:12), ncol = 4, byrow = TRUE)
summary(egII.nb)

str(cell2nb(nrow = 3, ncol = 4, type = "queen"))

egII.lw = nb2listw(egII.nb) ## Spatial weights (row standardized)
egII.lw$weights

nb2listw(egII.nb, style = "B")$weights # Binary weight coding (0/1)

nb2listw(egII.nb, style = "C")$weights # Globally standardized

nb2listw(egII.nb, style = "U")$weights # Globally standardized (unity)
```

Outline

1. Lattice Data-From Neighbors to Statistics
2. Introduction to Lattice Data
3. BW Joins/Test I
4. R-Demo
5. Assignment 1

Assignment 1

- **Problem 1 (15 pts):** Bowhead whale locations on a 4×6 lattice

$$\begin{array}{cccccc} 1 & 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 2 & 2 & 1 & 2 & 0 & 3 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{array}$$

  Recode nonzero cells as black and zeros as white. Using `joincount.multi()`, conduct z-tests for H_0 : CSR vs. H_A : not CSR based on BB, BW, and WW joins under rook neighbors with binary weights (`style="B"`). Report z-statistics and p-values. Then interpret the results and describe the spatial pattern.

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- **Problem 2 (10 pts):** Repeat Problem 1 using queen neighbors. Comment on differences from the rook-neighbor results.

- **Problem 3 (5 pts):** By hand, list the first-order neighbors for each cell in mosaic pattern II and compare with `str(egII.nb)`. Then repeat for `type="queen"`. Submit your hand calculations as an image.